

* Surds Indices - 1:

• Laws of Indices:

1) $a^m \times a^n = a^{m+n}$ eg. $2^3 \times 2^7 = 2^{3+7}$

2) $\frac{a^m}{a^n} = a^{m-n}$ eg. $\frac{2^5}{2^2} = 2^{5-2}$

3) $a^m \times b^m = (a \times b)^m$ eg. $3^2 \times 2^2 = (3 \times 2)^2$

4) $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$ eg. $\frac{3^2}{2^2} = \left(\frac{3}{2}\right)^2$

5) $a^0 = 1$ eg. $1234^0 = 1$

6) $\sqrt[m]{a} = a^{\frac{1}{m}}$ eg. $\sqrt[3]{8} = 8^{\frac{1}{3}}$

7) $a^{-m} = \frac{1}{a^m}$ eg. $10^{-5} = \frac{1}{10^5}$

8) $(a^b)^c = a^{b \times c}$ eg. $(2^3)^4 = 2^{3 \times 4} = 2^{12}$

9) $a^b \neq a^{b \times c}$ eg. $2^3 \neq 2^{3 \times 4} \neq 2^{12}$

10) $a^m = a^n \rightarrow m=n$
Imp $\downarrow$
 $2^{3 \times 3 \times 3 \times 3} \Rightarrow 2^{81}$

$\sqrt[2]{16} \rightarrow$ Square root of 16

$\sqrt[3]{64} \rightarrow$ cube root of 64

$\sqrt[4]{256} \rightarrow$ 4th root of 256

$\sqrt[5]{32} \rightarrow$ 5th root of 32

$\sqrt[m]{a} \rightarrow$ mth root of a

Eg-1) $(2^3)^{2^4} \rightarrow 2^{3^2 \times 2^4}$

$\rightarrow 2^{9 \times 16}$

$\rightarrow 2^{144}$

$\rightarrow 2^{144}$

2) $(2^4)^{2^3} \rightarrow 2^{4 \times 8}$
 $\rightarrow 2^{32}$

3) $(2^3)^{2^4} \rightarrow 2^{9 \times 4}$
 $\rightarrow 2^{36}$

4) $(2^2)^{2^3} \rightarrow 2^{8 \times 9}$
 $\rightarrow 2^{72}$

5) $(2^2)^{2^2} \rightarrow 2^{2 \times 4}$
 $\rightarrow 2^8$

6) $2^{2^2} \rightarrow 2^{2 \times 2 \times 2 \times 2}$
 $\rightarrow 2^{16}$

7) $2^{2^8} \rightarrow 2^{256}$

8) $2^{4^3} \rightarrow 2^{4 \times 4 \times 4}$
 $\rightarrow 2^{64}$

9) $2^{3^4} \rightarrow 2^{3 \times 3 \times 3 \times 3}$
 $\rightarrow 2^{81}$

10) $(2^4)^{2^5} \rightarrow 2^{4 \times 32}$
 $\rightarrow 2^{128}$

• Box-1

$$x^2 = 25$$

$$x = \pm 5$$

• Box-2 (Imp)

$$x = \sqrt{25}$$

$$x = 5$$

• Box-3

$$\sqrt{x} = 16$$

$$x = 256$$

• Box-4

$$\sqrt[3]{y} = 8$$

$$y = 512$$

• Box-5

(Imp)

$$x^2 = 4 \rightarrow -2, +2$$

$$x^3 = 8 \rightarrow 2$$

$$x^4 = 16 \rightarrow -2, +2$$

$$x^5 = 32 \rightarrow 2$$

$$x^6 = 64 \rightarrow -2, +2$$

Even power = $\pm$ ✓

odd power = $+$ ✓

• Box-6

$$A^2 = 4 \rightarrow \pm 2$$

$$B^3 = 64 \rightarrow 4$$

$$C = \sqrt{64} \rightarrow 8$$

$$D = \sqrt[3]{64} \rightarrow 4$$

$$\sqrt{E} = 4 \rightarrow 16$$

$$\sqrt[3]{F} = 8 \rightarrow 512$$

• Box-7

$$x^{1/2} \rightarrow \sqrt{x}$$

$$y^{1/3} \rightarrow \sqrt[3]{y}$$

Imp. Box-8

$$x = \sqrt{x} \sqrt{x}$$

$$z = \sqrt{2} \sqrt{2}$$

* Home Work :-

$$1) K^2 = 16$$

$$K = \pm 4$$

$$2) K^3 = 64$$

$$K = 4$$

$$3) \sqrt{K} = 4$$

$$K = 16$$

$$4) \sqrt[3]{K} = 8$$

$$K = 512$$

$$5) K = \sqrt{16}$$

$$K = 4$$

$$6) K = \sqrt[3]{16}$$

$$7) K^2 - 64 = 0$$

$$K = \pm 8$$

$$8) K^3 - 8 = 0$$

$$K = 2$$

$$\text{Imp } 9) 16 = \sqrt{K}$$

$$K = 256$$

$$\text{Imp } 10) 8 = K^{\frac{1}{3}}$$

$$K = 512$$

$$11) K^{\frac{1}{2}} = 36$$

$$K = 1296$$

$$12) K^{\frac{1}{3}} - 8 = 0$$

$$K = 512$$

$$13) (5^2)^3 = 5^K$$

$$14) 5^2^3 = 5^K$$

$$15) (7)^K = (7^2)^{3^2}$$

$$5^6 = 5^K$$

$$K = 6$$

$$5^8 = 5^K$$

$$K = 8$$

$$7^K = 7^{4 \times 9}$$

$$7^K = 7^{36}$$

$$K = 36$$

$$16) (11)^K = (11^5)^{2^3}$$

$$11^K = 11^{5 \times 8}$$

$$11^K = 11^{40}$$

$$K = 40$$

Imp

$$17) \frac{x^{4/3}}{8} - \frac{2}{x^{2/3}} = 0$$

$$\frac{x^{4/3} \times x^{2/3} - 16}{8x^{2/3}} = 0$$

$$\cancel{x^{4/3}} = 16$$

$$x^4 = 16^3$$

$$\cancel{x^4 = 4096}$$

$$x^{6/3} - 16 = 0$$

$$x^2 = 16$$

$$x = \pm 4$$

$$18) \frac{y^{7/5}}{4} - \frac{2}{y^{8/5}} = 0$$

$$y^{7/5} \times y^{8/5} - 8 = 0$$

$$y^3 = 8$$

$$\boxed{y=2}$$

$$\frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} = \frac{1}{2}$$

$$19) 5^{\overset{III}{7}} = 5^K$$

$$\rightarrow 5^7 = 5^K$$

$$K = 7$$

$$20) \frac{K^{1/6}}{4} - \frac{1}{K^{1/3}} = 0$$

$$K^{1/6} \times K^{1/3} - 4 = 0$$

$$K^{1/2} = 4$$

$$K = \boxed{16}$$

$$21) \frac{K^{1/6}}{8} - \frac{1}{K^{1/6}} = 0$$

$$K^{1/3} = 8$$

$$K = 512$$

$$22) \frac{5^9 \times 25^7 \times 125^8}{625^{11}} = 5^K$$

$$\frac{5^8 \times 5^1 \times 25^7 \times 125^8}{625^{11}}$$

$$\frac{625^3 \times 25 \times 5^8 \times 125^8}{625^3 \times 5^8 \times 125^8}$$

$$= 5^2 \times 5^1$$

$$= 5^3 = 5^K$$

$$K = \boxed{3}$$

$$23) \frac{6^{-9} \times 216^5}{36^{-4}} = 6^K$$

$$\Rightarrow \frac{6^{-9} \times \cancel{6 \times 6 \times 6} \times 36 \times 5}{\cancel{6} \times 6^{-2} \times 6^{-2}}$$

$$\Rightarrow \frac{6^{-9} \times 36^5 \times 6^5}{36^{-4}}$$

$$\Rightarrow \frac{6^{-9} \times 36^5 \times 6^5}{(6^2)^{-4}}$$

$$= \frac{6^{-9} \times 36^5 \times 6^5}{6^{-8}}$$

$$\Rightarrow 6^{-1} \times 6^5 \times 36^5$$

$$\Rightarrow 6^4 \times 36^5$$

$$\Rightarrow 6^4 \times (6^2)^5$$

$$\Rightarrow 6^4 \times 6^{10}$$

$$= 6^{14}$$

$$= 6^{14} = 6^K$$

$$K = \boxed{14}$$

$$24) \frac{9^{11} \times 27^{-2} \times 81^{\frac{7}{8}}}{243^5 \times 3^9} = 3^K$$

$$\frac{(3^2)^{11} \times (3^3)^{-2} \times (3^4)^{\frac{7}{8}}}{(3^5)^5 \times 3^9} = 3^K$$

$$= \frac{3^{22} \times 3^{-6} \times 3^{\frac{28}{8}}}{3^{25} \times 3^9}$$

$$\Rightarrow 3^{-6} \times 3^3 \times 3^{14} = 3^K$$

$$\Rightarrow 3^3 \times 3^8 = 3^K$$

$$3^{11} = 3^K$$

$$\boxed{K = 11}$$

$$25) \frac{2^9 \times 4^{-3} \times 8^7}{16^{-5} \times 32^{-4}} = 2^{K+10}$$

$$\frac{2^9 \times 2^{-6} \times 2^{21}}{2^{-20} \times 2^{-20}} = 2^{K+10}$$

$$\frac{2^{24} \times 2^{40}}{2^{40}} = 2^{K+10}$$

$$2^{64} = 2^{K+10}$$

$$K+10 = 64$$

$$\boxed{K = 54}$$

$$26) \frac{5^{-4} \times 125^9 \times 25^5}{3125^{-7} \times 625^{-4}} = 5^{\left(\frac{K}{3}-4\right)}$$

$$\frac{5^{-4} \times 5^{27} \times 5^{10}}{5^{-35} \times 5^{-16}} = 5^{\left(\frac{K}{3}-4\right)}$$

$$5^{37} \times 5^{35} \times 5^{12} = 5^{\left(\frac{K}{3}-4\right)}$$

$$5^{84} = 5^{\left(\frac{K}{3}-4\right)}$$

$$\frac{K}{3} - 4 = 84$$

$$\boxed{K = 270}$$

$$\frac{K}{3} = 73$$

$$\frac{K}{3} - 4 = 84$$

$$K = 88 \times 3$$

$$K = 264$$

$$27) \frac{16^{-9} \times 4^5 \times 2^6}{8^3 \times 64^{-5} \times 32^7} = 2^{\left(\frac{K+5}{2}\right)}$$

$$\frac{2^{-36} \times 2^{10} \times 2^6}{2^9 \times 2^{-30} \times 2^{35}} = 2^{\left(\frac{K+5}{2}\right)}$$

$$\frac{2^{-6} \times 2^6 \times 2^1}{2^{35}} = 2^{\left(\frac{K+5}{2}\right)}$$

$$\frac{2^1 \times 2^{35}}{2^{-34}} = 2^{\left(\frac{K+5}{2}\right)}$$

$$2^{-34} = 2^{\left(\frac{K+5}{2}\right)}$$

$$\frac{K+5}{2} = -34$$

$$K = -68 - 5$$

$$K = -73$$

$$28) \frac{13'' \times 4^8 \times 64^{-7} \times 6''}{16^{-5} \times 39'' \times 256^4} = K$$

$$\rightarrow \frac{13'' \times 2^{16} \times 2^{-42} \times 6''}{2^{-20} \times 39'' \times 2^{32}} = K$$

$$\rightarrow \frac{13'' \times 2^{-22} \times 6''}{39'' \times 2^{16}} = K$$

$$\rightarrow \frac{13'' \times 2^{-22} \times 6''}{13'' \times 3'' \times 2^{16}} = K$$

$$\rightarrow \frac{2^{-22} \times 2'' \times 3''}{3'' \times 2^{16}} = K$$

$$\rightarrow \frac{2^{-11}}{2^{16}} = K$$

$$\rightarrow 2^{-11} \times 2^{-16} = K$$

$$\rightarrow 2^{-27} = K$$

$$K = 2^{-27}$$

$$29) \frac{8^5 \times 6^5 \times 64^{-3} \times 16^7}{64^{-11} \times 4^8 \times 12^5} = K$$

$$\rightarrow \frac{2^{15} \times 2^5 \times 3^5 \times 2^{28}}{64^{-8} \times 2^{16} \times 3^5 \times 4^5} = K$$

$$= \frac{2^{15} \times 2^5 \times 3^5 \times 2^{28}}{2^{-48} \times 2^{16} \times 3^5 \times 2^{10}} = K$$

$$= \frac{2^{15} \times 2^5 \times 2^{28}}{2^{26} \times 2^{-48}} = K$$

$$= \frac{2^{15} \times 2^5 \times 2^{28}}{2^{-22}} = K$$

$$= 2^{15} \times 2^5 \times 2^{28} \times 2^{22} = K$$

$$\rightarrow \begin{array}{l} 2^{70} = K \\ \boxed{K = 2^{70}} \end{array} \rightarrow \begin{array}{l} \left(\frac{2}{2} \right)^{35} \\ \rightarrow 4^{35} \\ = \end{array}$$

$$30) \frac{16^9 \times 4^{-7} \times 64^5}{256^8 \times 1024^{11}} = 2^{2K}$$

$$\frac{2^{36} \times 2^{-14} \times 2^{30}}{2^{64} \times 2^{110}} = 2^{2K}$$

$$\frac{2^{66} \times 2^{-14}}{2^{64} \times 2^{110}} = 2^{2K}$$

$$2^{-12} \times 2^{-110} = 2^{2K}$$

$$2^{-122} = 2^{2K}$$

$$2K = -122$$

$$\boxed{K = -61}$$